

Association of stromal tumor-infiltrating lymphocytes with clinicopathological parameters in endometrial cancer

Sana Ahuja, Ashi Dubey, Sufian Zaheer

Department of Pathology, Safdarjung Hospital, Vardhman Mahavir Medical College, New Delhi, India

Purpose: Endometrial cancer (EC) ranks as one of the most prevalent gynecological malignancies globally. The presence and role of stromal tumor-infiltrating lymphocytes (TILs) in the tumor microenvironment have garnered interest due to their prognostic and therapeutic potential. This study aimed to evaluate the association between stromal TILs and various clinicopathological parameters in EC.

Methods: A prospective study was conducted which included 30 histologically confirmed cases of endometrial carcinoma. Specimens collected from January 2023 to June 2024 were processed for routine histopathological examination and immunohistochemistry for CD3 and CD20 markers. TILs were quantified as per the International Immuno-Oncology Biomarker Working Group guidelines and categorized into low (<20%) and high (≥20%) TILs.

Results: The study comprised 30 female patients, predominantly aged 51 to 60 years. Most tumors were of the endometrioid subtype (93.3%). High TILs were significantly associated with early tumor stage, lower grade, lesser myometrial invasion, and absence of nodal involvement on univariate analysis and with lower tumor stage and grade on multivariate analysis. No significant association was found between TILs and age, lymphovascular, or perineural invasion.

Conclusion: The findings suggest that high TIL infiltration correlates with favorable tumor characteristics, potentially serving as a prognostic marker for early and less aggressive EC. High TILs were associated with better tumor stage, grade, and reduced nodal involvement, indicating their protective role in tumor progression. However, the lack of association with certain parameters calls for further investigation into the functional state of TILs and their interactions within the tumor microenvironment.

Keywords: Tumor infiltrating lymphocytes, Adenocarcinoma, Endometrioid, Endometrial neoplasms

INTRODUCTION

Endometrial cancer (EC) is one of the most common gynecological malignancies, with increasing incidence rates worldwide. Based on GLOBOCAN (global cancer statistics), there were approximately 417,367 new cases of EC and 97,370 related deaths globally in 2020. Furthermore, in 2020, EC ranked as the second most prevalent gynecological cancer and was the third leading cause of death among women due to gynecological cancers worldwide [1].

The tumor microenvironment, particularly the presence and role of stromal tumor-infiltrating lymphocytes (TILs), has gained significant interest in recent years due to its potential prognostic and therapeutic implications. TILs, which include various subsets of immune cells such as T lymphocytes, B lymphocytes, and natural killer cells, have been studied extensively in different cancers like breast, head and neck squamous cell carcinoma, colorectal and lung cancer and are associated with improved prognosis and better clinical outcomes [2-5]. However, there is limited data on the association of TILs with clinicopathological parameters in EC.

This study evaluated the association between stromal TILs and clinicopathological parameters in EC.

METHODS

Study design and ethical considerations

The study was done in accordance with the Helsinki Declaration. The study was approved by the Institutional Ethics Committee of

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Correspondence to: Sufian Zaheer

Department of Pathology, Safdarjung Hospital, Vardhman Mahavir Medical College, Ansari Nagar, New Delhi 110029, India

Tel: +91-9650594818, **Fax:** +91-9958188632

E-mail: sufianzaheer@gmail.com

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Safdarjung Hospital (IEC/VMMC/SJH/02-2024/242). Written informed consent was taken from all patients included in the study.

Inclusion and exclusion criteria

All resection specimens of histologically confirmed cases of endometrial carcinoma were included, and cases that had received any neoadjuvant chemotherapy or radiation therapy prior to surgery were excluded from the study.

Sample collection and processing

The resected specimens of endometrial carcinoma received in the histopathology laboratory over 18 months from January 2023 to June 2024 were considered for the study. Clinical details such as age, gender, tumor grade, and stage were retrieved from the requisition forms. A total of 30 cases that met the inclusion criteria were selected for the study. Routine histopathological processing was performed, followed by hematoxylin and eosin (H&E) staining and immunohistochemistry for CD3 and CD20 to identify and quantify TILs.

TILs quantification

TILs were quantified using the method recommended by the In-

ternational Immuno-Oncology Biomarker Working Group for the standardization of TIL assessment in H&E-stained sections. The entire slide was initially scanned at low magnification ($\times 5$ or $\times 10$ objective lens), followed by a higher magnification ($\times 20$ objective lens) for detailed evaluation. Stromal TILs were defined as the percentage of the stromal area occupied by infiltrating lymphocytes. The average number of TILs was assessed in multiple stromal areas, excluding areas of necrosis, fibrosis, crush artifacts, and abscess formation. Additionally, TILs in stromal areas not adjacent to the tumor were excluded [6]. TILs were recorded as percentages and categorized into low TILs ($< 20\%$) and high TILs ($\geq 20\%$). Pathologists were blinded to the clinicopathological characteristics during TIL quantification.

Statistical analysis

Statistical analysis was conducted using the SPSS software, version 25.0 (IBM Corp.). The Fisher exact and chi-square tests were utilized to compare clinicopathological variables with TIL levels. Multivariate logistic regression was used to exclude the effect of confounding variables. A P-value of less than 0.05 was considered statistically significant.

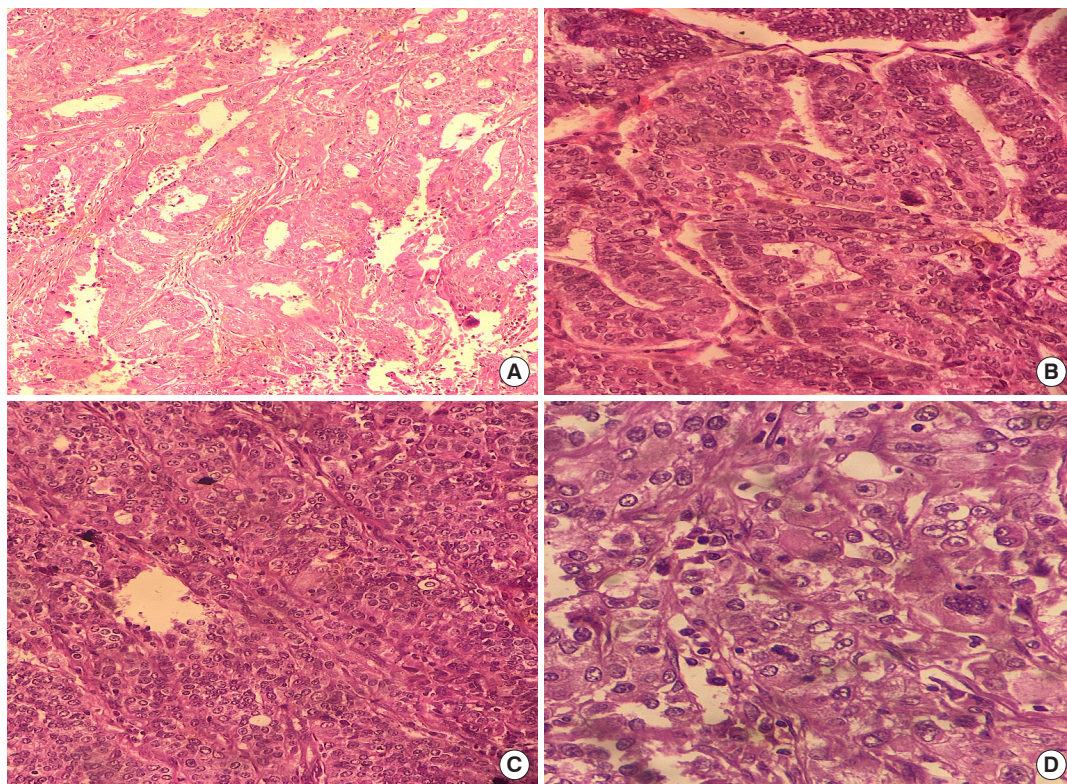


Fig. 1. Tumor grade in endometrial carcinoma. (A, B) A case of well-differentiated adenocarcinoma (hematoxylin and eosin staining, $\times 100$). (C, D) A case of poorly differentiated adenocarcinoma (hematoxylin and eosin staining, $\times 400$).

RESULTS

The study included 30 cases of endometrial carcinoma. The majority of patients ($n = 18$, 60%) were in the age group of 51–60 years, followed by 41–50 years ($n = 8$, 26.7%) and above 60 years ($n = 4$, 13.3%). The mean age of the patients was 54.2 ± 7.1 years. All patients were female.

Histologically, most tumors were of the endometrioid subtype ($n = 28$, 93.3%), followed by serous ($n = 2$, 6.7%). Tumor grading revealed that 12 cases (40%) were grade 1, 13 cases (43.3%) were grade 2, and five cases (16.7%) were grade 3 (Fig. 1).

Based on tumor stage, 11 (36.7%), 11 (36.7%), five (16.7%), and three (10%) cases belonged to pT1a, pT1b, pT2 and pT3 respectively. None of the cases belonged to T4. According to the nodal status, three cases (10%) showed N1 stage while 27 cases (90%) had no nodal involvement. None of the cases had N2 or N3 stage. Out of 30 cases, the lymphovascular invasion was observed in five cases (16.7%). None of the cases showed perineural invasion. Based on the categorization of TILs into low and high, 10 cases (33.3%) exhibited a low infiltration while the remaining 20 cases (66.7%) showed a high infiltration of TILs (Fig. 2).

Table 1 depicts the association of TILs with clinicopathological parameters. Stromal TILs had a significant association with tumor stage ($P = 0.04$), grade ($P = 0.01$), depth of myometrial involvement ($P = 0.03$), and nodal involvement ($P = 0.01$) on univariate analysis. However, no significant association was found with age, lymphovascular and perineural invasion. Multivariate logistic regression showed a significant association of high TILs with lower stage ($P = 0.03$) and grade ($P = 0.04$). Immunohistochemistry showed an almost similar proportion of CD3 and CD20 positive cells in all the cases and showed no significant association with clinicopathological parameters.

DISCUSSION

The study aimed to evaluate the association between stromal TILs and clinicopathological parameters in EC. Our findings reveal significant associations between TIL levels and several clinicopathological parameters, which underscores the potential prognostic value of TILs in endometrial carcinoma.

The majority of the patients in our study were in the age group of 51 to 60 years, consistent with the known epidemiological data

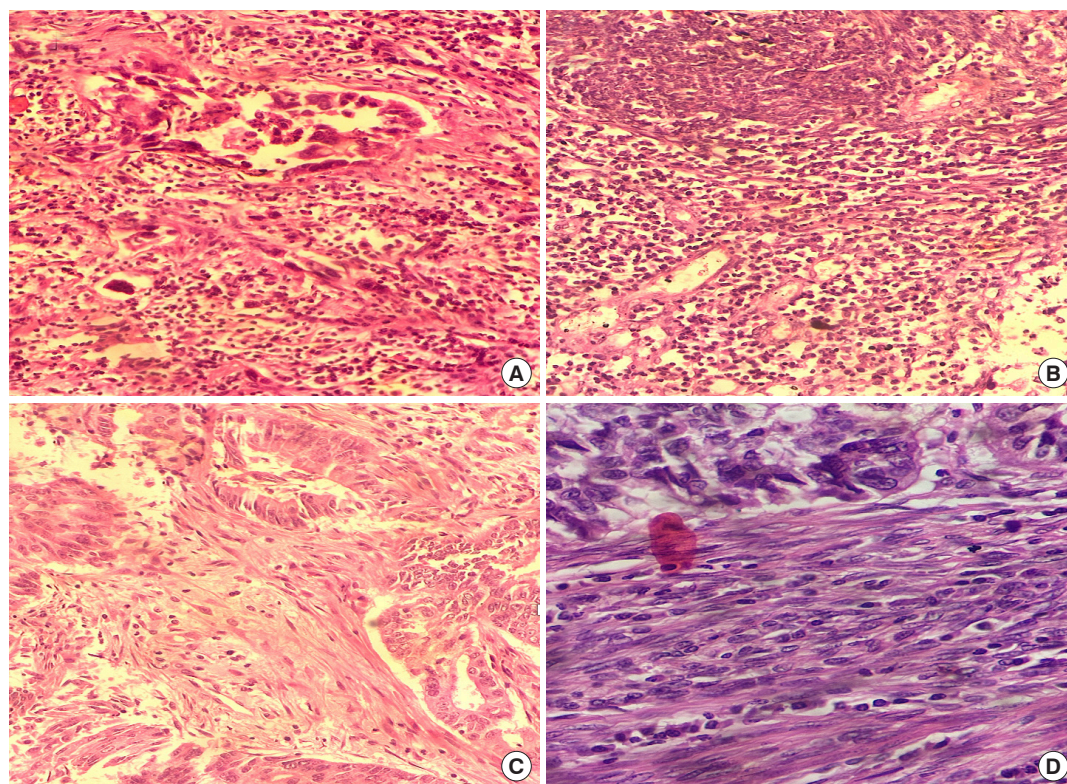


Fig. 2. Quantification of tumor-infiltrating lymphocytes (TILs) in endometrial carcinoma. (A, B) In the case of moderately-differentiated adenocarcinoma ($\geq 20\%$ TILs): 70% to 80% of TILs. (C, D) In the case of well-differentiated adenocarcinoma ($< 20\%$ TILs): 10% of TILs. Hematoxylin and eosin staining ($\times 100$).

Table 1. Association between clinicopathological parameters with TILs

Parameter	TILs		Univariate analysis P-value	Multivariate analysis P-value
	Low (n = 10)	High (n = 20)		
Age			0.52	0.91
≤ 60 yr (n = 26)	8	18		
> 60 yr (n = 4)	2	2		
Tumor stage			0.04	0.03
T1 (n = 22)	5	17		
T2-T3 (n = 8)	5	3		
Nodal involvement			0.01	0.63
N0 (n = 25)	6	19		
N1-3 (n = 5)	4	1		
Grade			0.01	0.04
G1 (n = 12)	1	11		
G2 (n = 13)	5	8		
G3 (n = 5)	4	1		
Myometrial invasion			0.03	0.42
< 50% (n = 11)	1	10		
≥ 50% (n = 19)	9	10		
Lymphovascular invasion			0.24	0.81
Present (n = 3)	2	1		
Absent (n = 25)	8	19		

TILs, tumor-infiltrating lymphocytes.

indicating higher incidence rates of EC in postmenopausal women [7]. The predominance of the endometrioid subtype (93.3%) in our cohort aligns with global trends, where endometrioid adenocarcinoma is the most common histological subtype of EC. Tumor grading and staging data from our study were also in line with existing literature, where early-stage and lower-grade tumors are more prevalent [7].

Our results demonstrated a significant association between high TIL levels and early tumor stage ($P = 0.04$), lower grade ($P = 0.01$), and lesser depth of myometrial invasion ($P = 0.03$). Multivariate analysis showed a significant association with lower stage and grade, reinforcing the observation that higher TIL infiltration is linked with more favorable tumor characteristics. These findings suggest that higher TIL infiltration is associated with more favorable tumor characteristics, which may correlate with a better prognosis. Specifically, the majority of T1 stage and G1 grade tumors exhibited high TIL levels, indicating that stromal TILs could serve as a marker for early and less aggressive disease. Ozturk et al. [7] also observed high TILs to be associated with International Federation of Gynecology and Obstetrics (FIGO) grade 1 tumors, lower nuclear grade, early pathological stage, small size, myometrial invasion less than 50% and improved overall survival.

Fan et al. [8] observed improved progression-free survival in associated with TILs in high grade endometrial carcinoma thus suggesting a prognostic role. Kono-Sato et al. [9] concluded that low CD3+ TILs and low CD8+ TILs were independent worse prognostic factors. These studies could not find any association with grade, stage, lymphovascular invasion, or myometrial invasion.

Bounous et al. [10] observed that high TILs, did not show a significant statistical correlation with histology, tumor grade, or FIGO stage. However, mismatch repair deficient tumors demonstrated a higher mean value of TILs than the other tumors.

Mendiola et al. [11] demonstrated cytotoxic T cells (CD25+, PD-1+, and PD-L1+) showed a strong association with longer relapse-free survival. Further, they observed that a high stromal T-cell fraction of CD3+ PD-1+ cells was associated with a good 5-year relapse-free survival rate as compared to low CD3+ PD-1+ fraction.

Interestingly, there was also a significant association between high TIL levels and absence of nodal involvement ($P = 0.01$), suggesting that TILs might play a role in preventing tumor metastasis to lymph nodes. This reinforces the concept of TILs having a protective role in tumor progression, as observed in other cancers like breast, colorectal, and lung cancer.

Contrarily, no significant association was found between TIL levels and age, lymphovascular invasion, or perineural invasion. This indicates that while TILs are associated with certain tumor characteristics, their presence does not correlate with all clinicopathological parameters.

Immunohistochemical analysis revealed no significant difference in the proportion of CD3 and CD20 positive cells across cases, indicating a consistent presence of T lymphocytes and B lymphocytes among the study population. This uniformity suggests that the immune cell composition of TILs in EC may not vary significantly with clinicopathological factors, although their overall quantity does.

The limitations of the present study include a small sample size which may limit the generalizability of the findings. Further, it is a single institution study which could introduce a selection bias. Also, it primarily focused on the quantity of TILs, without evaluating their functional characteristics, which could provide further insights.

In conclusion, this study highlights the significant association of stromal TILs with tumor stage, grade, myometrial invasion, and nodal involvement in EC. These findings suggest that TILs could serve as potential prognostic markers, aiding in the stratification of patients based on their tumor's immunological profile. The assessment of TILs, as recommended by the International Immuno-Onco-logy Biomarker Working Group, is a straightforward and reli-

able test. Morphological evaluation of TILs can yield important prognostic insights in cases of endometrial carcinoma and could be integrated into routine histopathological reporting. However, the lack of association with age, lymphovascular invasion, and perineural invasion indicates that further research is needed to fully elucidate the role of TILs in EC. Future studies with larger cohorts and prospective designs are warranted to validate these findings and to explore the underlying mechanisms through which TILs influence tumor behavior. Additionally, investigating the functional states of TILs and their interaction with other components of the tumor microenvironment could provide deeper insights into their prognostic and therapeutic potential in EC.

CONFLICT OF INTEREST

No potential conflict of interest relevant to this article was reported.

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ORCID

Sana Ahuja <https://orcid.org/0000-0002-4630-016X>
 Ashi Dubey <https://orcid.org/0009-0007-4619-8680>
 Sufian Zaheer <https://orcid.org/0000-0002-8044-4327>

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